

Nishan Kharel

AI Engineer & Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with **1+ year of hands-on experience** building machine learning, NLP, and computer vision systems. Proven track record delivering end-to-end AI pipelines from data ingestion to production APIs using Python, SQL, LLMs, RAG, and modern deep learning frameworks. Adept at translating complex data problems into measurable business outcomes in recruitment, document intelligence, and real-world product contexts.

EDUCATION

Bachelor of Computer Science and Information Technology

March 2021 – Jan 2026

Texas International College, Tribhuvan University – Kathmandu, Nepal

TECHNICAL SKILLS

Languages	Python, SQL, C
ML & AI	Machine Learning, Deep Learning, NLP, Computer Vision, Fine-tuning, TensorFlow, OpenCV, Scikit-learn, Pandas, Jupyter
LLM / RAG / Vector	LLMs, RAG, pgvector, Sentence Transformers, OpenAI API, Gemini API, Prompt Engineering
Backend	FastAPI, Flask, SQLAlchemyORM, Alembic, SSL Server Configuration
Database	PostgreSQL, Vector Database
DevOps & Cloud	Docker and Docker Hub, Git
Dev Tools	Postman, Swagger, table-Plus, Docker
Soft Skills	Problem Solving, Communication, Teamwork, Analytical Thinking, Jira, Microsoft Project

PROFESSIONAL EXPERIENCE

AI Engineer (Part Time)

Oct 2025 – Feb 2026

Mantra Ideas Tech · Kathmandu, Nepal

- Engineered an **AI-based Bank Note OCR System** using computer vision and deep learning to detect and classify currency denominations, improving recognition accuracy and operational reliability.
- Fine-tuned **Text-to-Speech (TTS) models** for Nepali voice synthesis, optimising speaker adaptation and naturalness for real-time speech generation with VibeVoice and Voice Cloning.
- Built NLP automation tools including intelligent document parsing, information extraction pipelines, and API-integrated AI solutions to streamline recruitment and content processing workflows.

Data Scientist

Mar 2025 – Aug 2025

Mind Risers Technology · Kathmandu, Nepal

- Developed an **AI-powered ATS system** for Recruit Nepal using Tesseract OCR, spaCy NER, and semantic search to match candidate profiles with job descriptions, improving recruiter efficiency and user engagement.
- Built a **Nepali fake news classifier** for the Pop-Up app using BERT-based Transformer models, enabling real-time high-accuracy content verification.
- Created an **automated Job Description Generator** with secure API integration, enabling dynamic job-posting creation from role and organisational inputs.

Data Science Consultant (Part-time)

Feb 2025 – Feb 2026

Namaste Network · Chabahil, Kathmandu

- Designed and deployed end-to-end data science solutions covering predictive modelling, NLP pipelines, and automated analytics using Python and SQL.
- Delivered training on Neural Networks and Transformer architectures, covering backpropagation, attention mechanisms, and real-world NLP applications.
- Designed data science bootcamp curriculum and materials, **improving student satisfaction by 25%**.

Machine Learning Tutor

2024

Texas International College · Kathmandu, Nepal

- Conducted **20+ lectures and mentoring sessions** on AI and Machine Learning for undergraduate students.
- Facilitated hands-on ML workshops, achieving a **90% student pass rate**.

KEY PROJECTS

Bank Note OCR: Vector Similarity and Computer Vision

 [GitHub](#)

- Built a currency detection pipeline using computer vision and deep learning to accurately classify Nepali banknote denominations in real time.
- Applied vector similarity techniques for robust denomination matching under varied lighting and image quality conditions.

Recruit Nepal: AI-Powered Job Matching & ATS Platform

 [GitHub](#)

- Built an end-to-end ATS pipeline with Tesseract OCR fallback, spaCy NER for structured JSON extraction, and Sentence Transformer embeddings.
- Applied cosine similarity for Top-N candidate shortlisting followed by **LLM-based contextual ATS scoring** and automated candidate ranking.

Shishu Care: Infant Cry Classification System

 [GitHub](#)

- Designed a **CNN-BiLSTM with attention** architecture to classify infant cries (hunger, pain, burping, sleep) from mel spectrograms with near real-time inference.
- Performed full audio preprocessing: noise handling, waveform normalisation, and mel spectrogram extraction from raw .wav files.

RAG Chatbot with pgvector

 [GitHub](#)

- Built a production-ready RAG chatbot using **FastAPI + PostgreSQL (pgvector) + Gemini LLM API** with Sentence Transformer embeddings for cosine-similarity nearest-neighbour retrieval.
- Containerised with **Docker-Compose**; interactive Streamlit frontend for querying.

PopUp: Vendor-to-Vendor Conversational Chatbot

- Developed a multi-document chatbot with **LangChain + OpenAI API**, dynamic conversational forms, and session state management for accurate multi-turn vendor workflows.

Protein Function Prediction- HealthCare (CAFA-6)

 [GitHub](#)

- Predicted Gene Ontology (GO) labels from amino acid sequences using deep learning, with bioinformatics preprocessing, custom encoding, and class-weighted training.

CERTIFICATIONS

- **Python Programming** – Programiz Pro, 2024
- **NumPy for Data Science** – Programiz Pro, 2024
- **Structured Query Language (SQL)** – Programiz Pro, 2023

PROFESSIONAL REFERENCES

- **Sarthak Shrestha** –Senior Frontend Developer,Mantra Ideas Tech
+977 9808315780
- **Pukar Upreti** – Bunnings, Software Engineer,Australia
pukarupreti3@gmail.com,0467334451

ADDITIONAL QUALIFICATIONS

- **Hackathon Achievement:** Secured **2nd place** at the *Code for Change* event; also participated in Kings College Hackademia hackathon focused on AI and ML, demonstrating strong applied problem-solving skills under competitive conditions.
- Actively research and implement cutting-edge ML/AI models from academic papers, technical documentation, and real-world datasets.